

COM3021 – Data Science at Scale

Virtualisation and Containerisation

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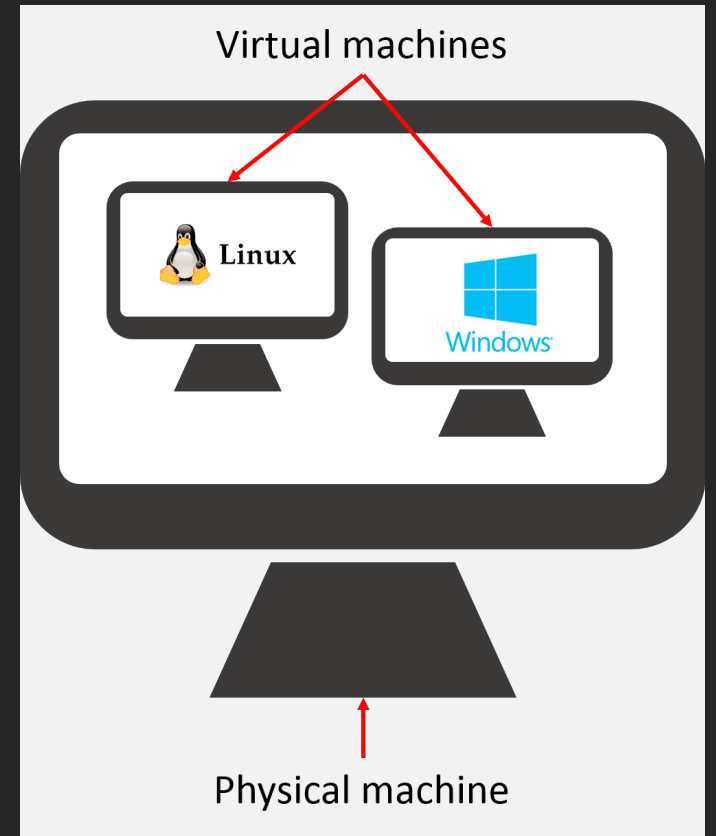


Introduction

- Virtualisation and containerisation both enable the abstraction of resources at different levels
 - Play an important role in distributed systems making it easier for managers to deploy updates
 - It improves the efficiency and availability of resources
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Virtualisation

- It is the creation of a software-based *virtual machine* that acts like a real computer with all of its components (hardware, peripherals, operating system etc).
- A virtualisation software running on the physical computer (host) allowing for one or more virtual computers (guests) to run independently from one another

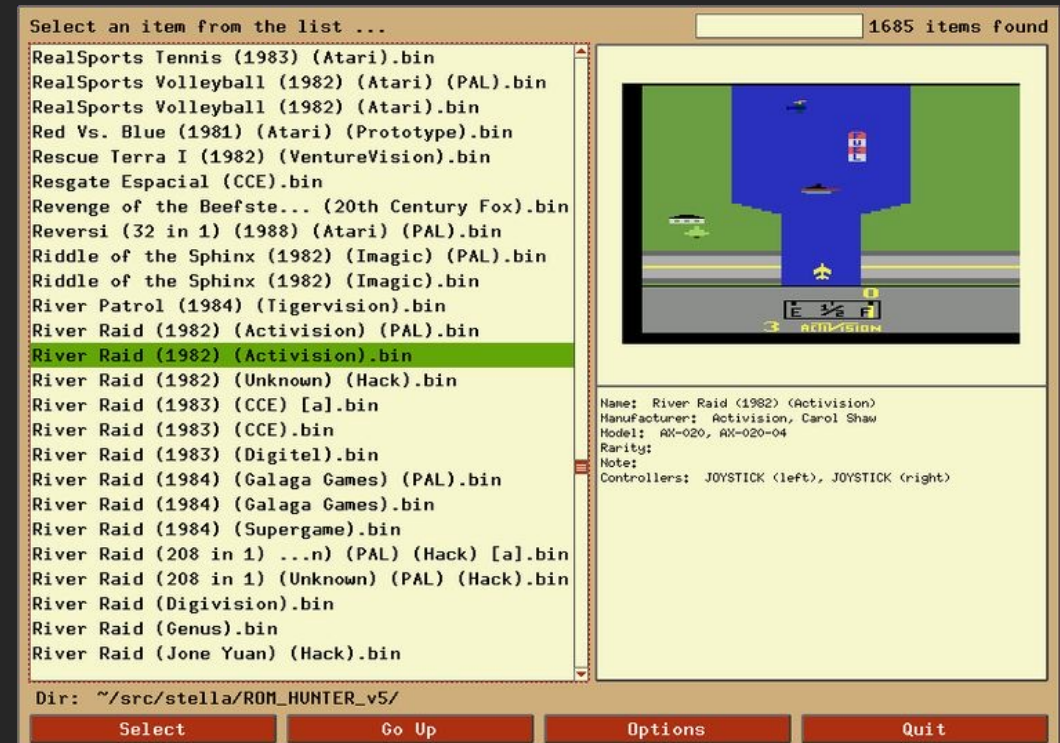


Virtualisation

- This specialised software is called hypervisor, virtual machine monitor or virtualiser.
 - There are many virtualisation softwares on the market
 - *VMWare Workstation (VMWare)*
 - *VirtualBox (open-source but maintained by Oracle)*
 - *Hyper-V (Microsoft)*
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Virtualisation

- Not the same thing as emulation
 - *An emulator is a software that mimics a specialised hardware allowing another hardware to execute its software*
 - *In an emulated environment, the guest software (e.g., a retro game) does not interact directly with the host hardware.*
 - *In a virtualised environment, the guest system will interact with the host hardware*



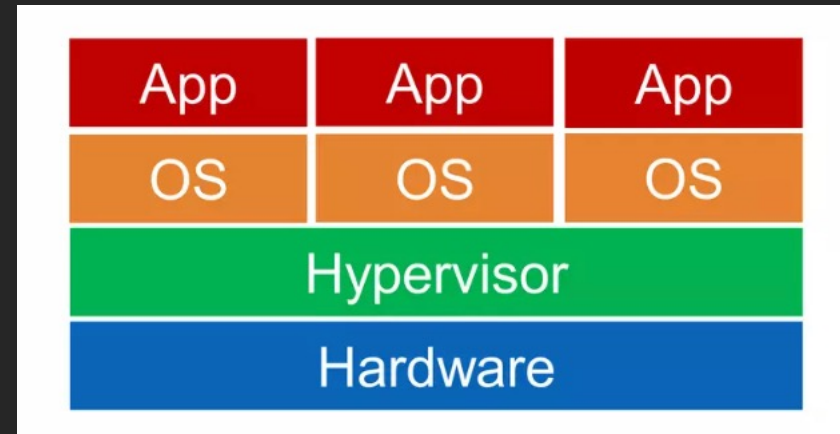
Snapshots

- A snapshot is a state of a virtual machine at an exact point in time.
 - A snapshot enables the virtual machine's state at the time of the snapshot to be restored later, effectively undoing any changes that occurred afterwards.
 - These snapshots can be copied and migrated over to other host machine
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Drawbacks

- Significant overhead
- Lots of redundancies

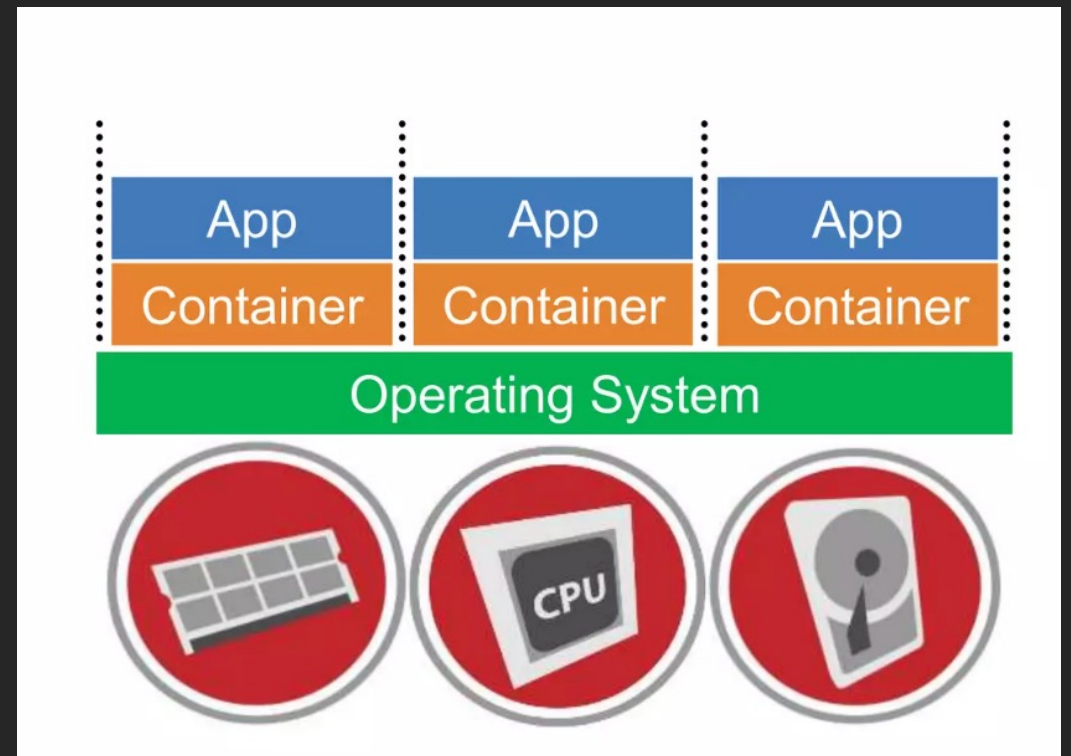
Each VM will have its own operating system, configuration files etc



Containerisation

Containerisation

- Containerisation (AKA OS-level virtualisation) enables the abstraction at the operating system level, allowing multiple applications to share binaries while remaining isolated from one another
- Containers can only use the resources that have been allocated to them



Containerisation

- A container is essentially a fully packaged and portable computing environment.
- Containers are meant to be:

portable

scalable

easier to build and deploy

easier to manage

isolated

Container orchestration

- Container orchestration automates the deployment, management, scaling, and networking of containers.
- Platform as a service (PaaS) is an enabler for software development where a third-party service provider delivers a platform to customers so they can develop, run, and manage software applications without the need to build and maintain the underlying infrastructure themselves.

